Ameya Panchal

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Education

The Pennsylvania State University University Park, PA

B.S. Computer Science, Minor in Physics, AI • Schreyer Honors College Expected May 2028

- GPA: 3.71 / 4.00
- Relevant Coursework: Linear Algebra, Data Structures & Algorithms, Systems Programming, Discrete Mathematics, Physics (Mechanics / E&M)

Research

Reading GPT-2's IOI Circuit Three Ways

Independent • TransformerLens Jul 2026

https://ameya-bit.github.io/posts/ioi-circuit/index.html

- Recovered the IOI circuit (Wang et al.) with direct logit attribution, activation patching, and path patching, each hand-implemented
- Framed as a methods comparison: DLA finds the name-mover heads, patching finds what feeds them, path patching isolates the link between — each blind to what the next reveals

An Induction Head in Disguise — Circuit Analysis of a Character-Level Transformer

Independent • PyTorch, TransformerLens • *LessWrong* Jul 2026

https://www.lesswrong.com/posts/x2ZdCoqAaLedpjDk3/an-induction-head-in-disguise-chasing-grammar-in-a-character

- Trained a 6-block character-level transformer and isolated a copying head via OV-circuit eigenvalues and copying scores
- *Disconfirmed* two of my own hypotheses — quotation and parenthesis grammar — with targeted QK attention-pattern controls
- Found the OV circuit encodes bracket completion while the QK circuit stays blind to it: a half-built mechanism. Full code published alongside the writeup

Monte Carlo Radiative Transfer — Neutron-Star Pulse Profiles

Penn State Multi-Campus REU • advised by Prof. Asif ud-Doula • Python Oct 2025 – Present

https://sites.psu.edu/mcreu/2026/07/21/where-the-error-hides-how-a-common-assumption-biases-pulse-fraction-for-neutron-stars/

- Built a Monte Carlo scattering atmosphere feeding a general-relativistic pulse-profile model (Schwarzschild light-bending, Doppler) to quantify the systematic from the isotropic-emission assumption in neutron-star mass–radius measurements
- Validated against Chandrasekhar's exact H-function (reduced χ^2 = 0.70) and Bogdanov et al. waveforms (0.11% peak error)
- Showed the systematic *routes* between pulsed fraction and waveform shape by hot-spot geometry and spin; first-author paper in preparation

Experience

Nittany AI Advance Internship Program State College, PA

Application Specialist • Client: Penn State Office of the Physical Plant Dec 2025 – Present

- Scoped ambiguous client requirements into technical documentation for the Docfinity systematic
- Prototyped evaluation protocols for generative and OCR-derived content, and drove requirements to consensus across a cross-functional team ahead of launch.

Penn State Department of Physics University Park, PA

Physics Learning Assistant • PHYS 211 Aug – Dec 2025

- Taught calculus-based mechanics in problem sessions, labs, and exam prep

Skills & Interests

ML & Research: Python, PyTorch, HuggingFace Transformers, TransformerLens, NumPy, Scikit-learn, Git

Interpretability: Transformer circuits, induction heads, OV/QK analysis, activation and path patching, direct logit attribution

Methods: Monte Carlo simulation, numerical validation, linear algebra

Also: Java, JavaScript / React, SQL, Flask